

Achieving Court Verifiability without Expert Knowledge while Maintaining Coercion Resistance- A (2 Devices) and (3+ Receipts) in-booth e-voting system

Abstract. In this paper, we propose an in-booth e-voting solution that allows voters to challenge their individual votes in courts and allows judges to verify by themselves without experts. The proposed system uses 2 devices; the first of which, authenticates the voter and prints a receipt with a unique cryptographic ID code (*HBID*) for each voter. The second is a voting device that permits successive multiple voting and prints a receipt for each voting attempt, then prints a sealing receipt for poll workers with enough information to identify the voter's final receipt; we combine the Benaloh challenge idea and the multiple voting option to deceive vote coercers/buyers and detect malicious voting devices at the same time. We advise to include “*Boycott*” & “*Reject All*” options to avoid absent voting and have a complete protection from coercers. We also hide the printed vote inside a simple function of the vote & *HBID* to protect from coercion by poll workers (who will take the distinguishing receipt); although it is supposed to be fast & easy for the average person to get the vote from the simple function in the receipt, it would be time consuming and noticeable if poll workers did it inside the poll station for all voters. We then publish per booth summary reports every fixed interval to serve as check points and to be used by RLAs and ZK queries as well. Finally, election result is published in a Public Bulletin Board as records (*booth, time, vote, no of votes*) to anonymize it from adversaries.

Keywords: E-Voting, tracking codes, vote coercion, vote buying, RLAs.

1 Motivation & Main Design Decisions

E-voting techniques provide many helpful tools for verifiability and fraud/manipulation detection; however, as [1] put it “*for voters that do not verify or whose verification failed, we make no guarantees*”. If non verifying voters have only themselves to blame, e-voting systems should provide solutions for failed verification also distinguishing it from false claims. In this paper, we target an e-voting system that, while still being able to detect manipulations and avoid vote buying/coercion, ***gives the opportunity for any voter, or a group of voters, to challenge the voting system in court***; a losing candidate can gather a sample of supporters and go to court to check their votes. Another issue is what the constitutional German court described in [2,3] about cryptographic proofs as being “*somewhat untenable as a verification*” since they require expert testimony; an argument that would be further emphasized in developing countries, [4], which scores less in both the availability of highly qualified cryptographic experts and public trust in voting systems. Our proposed design aims to achieve what we call complete and sound court verifiability (*a court being able to decide without expert knowledge*) by using receipts with tracking codes and public bulletin boards.

We preferred in-booth e-voting to avoid infrastructure deployment problems and/or cyber security risks; we use a first device¹ to authenticate voters without repetition; i.e., checking this authenticated ID has not been here before to avoid repeated or absent voting. The first device² is also responsible for generating and printing a per voter tracking code, we call *HBID*, that will appear in all receipts. We do not let the voting machine use the original vote ID, or generate the HBID itself, to distribute trust between 2 isolated machines; with other entities as detailed later in the paper.

The second voting machine deals only with *HBID* as the new identity after reading a printed passing approval generated by the first machine. *We introduce a modified Benaloh challenge [7,8] option* by allowing multiple intermediate voting, but with receipts, to both detect a malicious voting device (the device has no way of telling whether this receipt is final or not) and at the same time have an expired receipt to show any vote coercer and/or buyer; valid *voting options should include “boycott” and “reject all”* for complete protection from coercers.

To go around *the dilemma of deceiving adversaries while not deceiving judges*, we print (in different color to avoid mistakes) an extra sealing receipt for poll workers to keep (includes final time and number of multiple votes); the sealing receipt should be submitted in court to either assert or falsify a voter’s claim (could also be used for Risk Limiting Audits RLAs).

To waive the assumption of “trusted poll officers” *we hide the vote inside a simple function*; i.e., the second device does not print the clear vote, but a simple function of the vote and the tracking code. The function should be simple enough to be reversible in 1-2 mins by voters and judges, yet it would be time consuming and publicly noticeable if a poll worker dedicated 1-2 mins/voter to check the vote before allowing the voter to sign and leave³.

keeping the clear vote in the sealing receipt and letting voters throw it in a ballot box, without poll workers interference, might jeopardize both system usability and integrity; voters may throw a wrong receipt, or even not throw it at all, whether mistakenly or deliberately. In our proposed solution, ***the sealing receipt is the election authority’s responsibility***; if they claimed in court that it is missing, then they are the one to be blamed/doubted. If voters were to throw it in the ballot box, we have no way of telling who is malicious or responsible. Finally, proving non-voting starts with a SNARK supporting non-inclusion proofs, then follows a court convincing methodology as detailed in Appendix A.2 (***absent voting***).

¹ This could be done in different possible ways; for example, scanning IDs/electronic IDs, biometric checks or both. There are problems associated with biometric devices, like those happened in Afghanistan elections [5], but they are the most used to avoid absent voting from Africa [4] to India [6] and even in less crowded polls like UAE. Hence, we assume proper functionality and suffice with consistency checks for now.

² Both devices should be isolated (air gapped) with a removable memory card.

³ In most real-world cases, poll officers are assumed trusted; only audited by independent observers and/or candidates’ representatives. The linear function is just an extra barrier from coercion, we believe complete protection is not possible here; they can force you to show any receipts/QR-codes, they can stand behind you while voting, ...etc. *If you are counting on auditors’ presence, then why not count on them regarding the linear function in the receipt?*

2 System Architecture & Voting Steps

The architecture of the system is described in Fig.1, follows is a further elaboration.

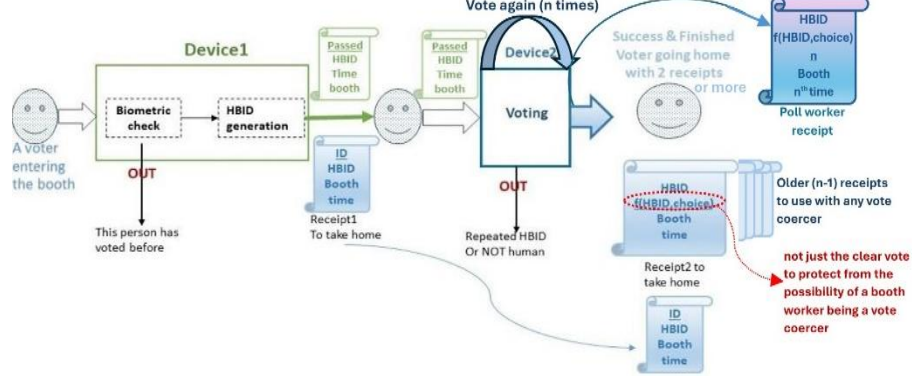


Fig.1. The voting process in the proposed system

-**The generated *HBID*** must be 1)hiding, 2)binding, and 3)unpredictable from the voter ID with overwhelming probability. If *HBIDs* were to have collisions the system would collapse; if voter ID could be revealed or predicted from it, coercers can code their calculations prior to election. Hence, it is a strong cryptographic hash function (possibly post-quantum) that includes randomization; same analogy when designing *verifiable random functions (VRFs)* [9].

-**The linear function printed in receipt2** holds a trade-off between usability and Poll officers' trust. It could be as simple as flipping a bit position in *HBID* that is function of the voter's choice (not the exact choice index to avoid border bits, say middle-choice index); i.e., if *HBID* is n bits and there are $m < n$ choices in this election, then $\text{Code}[i] = \text{HBID}[i]$; $\text{Complement}(\text{Code}[(n-m)/2 + \text{choice index}])$;

2.1 Voting Steps

1. When entering booth, the voter submits ID to device1 and authenticates he/she is a registered voter who hasn't voted yet⁴.
2. Device1 generates a new *HBID*, appends an encrypted tuple (ID, HBID, bio, time, booth) to its memory card and aggregates data (before encryption) in a Merkle/Verkle Tree.
3. Device1 prints a *timed passing ticket* to be scanned by device2, and a receipt1 for the voter to keep containing ID, HBID, Time, Booth. The passing ticket contains the *HBID* to be scanned by device2 to avoid human entering errors and attest passing the right to vote checks at the same time.

⁴ Example procedure could be to perform a biometric check, then starts some dialogue with the machine to prove humanity (to detect previously collected photos or any offline material inserted in the machine).

4. The voter goes to device2; submits his/her passing ticket; device2 checks that this HBID is not already in its memory⁵
5. Device2 shows the choices for the voter; the voter clicks on the desired choice, then device2 shows on screen a preview of a timed receipt with HBID and the simply coded choice $f(\text{choice}, \text{HBID})$; the print preview screen should show how $f(\text{choice}, \text{HBID})$ is calculated for the voter to understand the receipt meaning. If vote is confirmed as correct, device2 prints it and appends its info to the memory card inside it.
6. The voter may repeat step 5 several times, and every receipt will contain a vote counter. After confirming this is the final vote, device2 flags its record in memory and prints timed receipt3 to be handed to poll workers and kept in a box.

2.2 At Checkpoints and After Voting Closes

The following steps, as depicted in Fig.2, should take place after the voting closes; also, every fixed interval to detect and handle errors earlier.

7. Each device prints cryptographic checksums of its data for auditors and poll officers to check the consistency of data from both devices and consolidate them in a summary report. The report may contain the number of voters, no repeated HBIDs, Merkle/Verkle root of HBIDs, some more checks from device1 biometrics like no repetition, maybe Male/Female ratio. Then all of them sign the summary report either manually in a public document or by their public keys to a public Blockchain containing (**booth ID, date & time of session, no of voters, no of votes,...**); the number of voters should be also checked against handwritten signatures⁶ totals. Risk Limiting Audits (RLAs) can be performed by cryptographically verifying the correct corresponding data of sampled poll workers receipts in both devices⁷. If errors were found, they should be listed too.
8. Poll workers remove memory cards from both devices and deliver them along with the sealing receipts box to the election authority.
9. The election authority aggregates all memory cards in a voting server which will check IDs are not repeated in any other booth, match the 2 cards record by record⁸, check times are consistent, check consistency with subtotals in summary reports,...etc. We assume the remaining steps will be performed by the election authority as in most voting systems and deter the detailed analysis to future work.
10. When the voting server completes the counts, the election result is uploaded to a public bulletin board in the form (booth, time, choice, no of votes); we escape

⁵ It is possible to conduct a second humanity proof dialogue here to prevent automated voting.

⁶ So that, if the two machines colluded to have consistent fake summary reports, they will differ from the handwritten registry.

⁷ For example, checking that (time on device1 is < time on device2) with reasonable difference.

⁸ To avoid the possibility of an adversary getting access to the 2 cards (or their records) somehow and performs the matching as well to know the corresponding identity of each vote, each device maybe assigned an encryption key (by the election authority) to encrypt its record.

from vote buying/coercion depending on the fact that it is impossible for 2 votes to have the same exact time and booth. Hence, any voter can identify his/her vote in the Public Bulletin Board, while adversaries can at maximum figure the ratio of those who deceived them (can't find recorded time in the receipts they have).

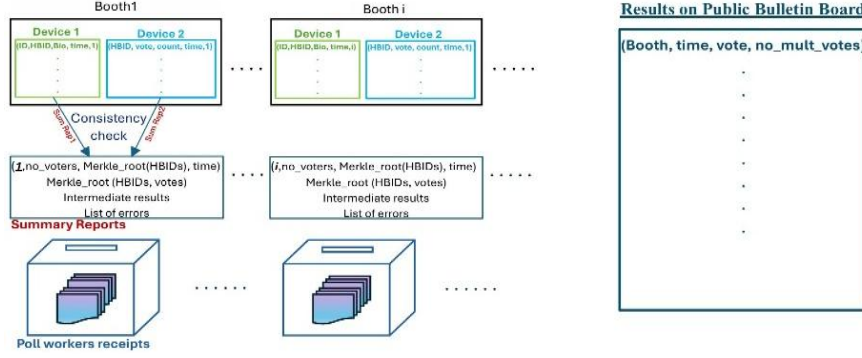


Fig.2. Data and Summary reports from booths after the election (or periodically)

3 Summary & Future Work

All reported statistics show that election QR codes are rarely verified ($\leq 10\%$), yet there is always a persistent minority of losing candidates who doubt the results and may challenge the election authority in court⁹. An honest judge is trapped between two fears; to be deceived by complicated terms out of his/her area of expertise, or to fall for hypothetical arguments and falsify the election (wasting money and effort) without solid proof.

In this paper, we proposed a new feature that even paper ballot voting systems lack; *court verifiability* is the ability of verifying individual votes in court. Then, we presented an E2E verified voting system that guarantees court verifiability through two final receipts that could be submitted in court and accepted as evidence. The system also allows expired voting receipts that could be used to deceive vote buyers/coercers; however, if a voter or a losing candidate tried to deceive the court with the same trick, the system provides an extra distinguishing receipt that is kept by the Election Authority and can be brought to court in dispute.

We tried to introduce a new middle point between traditional tradeoffs in e-voting systems that we believe is satisfying for some. Also, we proposed an interesting extra beneficial use to the classic Benaloh challenge. Appendix A discusses protocol assumptions, performance metrics, and robustness against different malicious components and/or manipulations.

Some details may have been abstracted like the hash function, the RLA technique used, and the exact consistency checks, also some could be altered according to system needs like the exact fields in the summary report; this perfectly fit automated formal verification as a future work analysis tool. Possible future work may also

⁹ That's why we do not consider *manipulation detection* as measured in [10] a problem here.

include formal proofs, deeper cryptanalysis and investigating coercion threats like those in [11]; comparisons with other systems that have some form of code trackers or dual paper ballots from the old Star vote [12,13] to newer ones [3,14].

References

1. Kristian Gjosteen et al., “Coercion Mitigation for Voting Systems with Trackers: A Selene Case Study”, E-Vote-ID 2023, LNCS14230, pp.69–86; https://doi.org/10.1007/978-3-031-43756-4_5
2. NDI: *The constitutionality of electronic voting in Germany*(2019), downloaded 1/12/2023; <https://www.oseinstitute.org/blog/electronic-voting-banned-in-germany>
3. Prashant Agrawal et al., “OpenVoting: Recoverability from Failures in Dual Voting”, E-Vote-ID 2023, LNCS14230, pp.18–34; https://doi.org/10.1007/978-3-031-43756-4_2.
4. Hina Bent Haq et al., “End-to-End Verifiable Voting for Developing Countries- What’s Hard in Lausanne is Harder Still in Lahore”, arXiv:2210.04764v1, [cs.CY], Oct 2022.
5. K. Panahi et al., “But they have overlooked a few things in Afghanistan: An Analysis of the Integration of Biometric Voter Verification in the 2019 Afghan Presidential Elections”, USENIX SS24, <https://www.usenix.org/conference/usenixsecurity24/presentation/panahi>
6. <https://www.brookings.edu/articles/indias-electoral-democracy-how-evms-curb-electoral-fraud/>, last accessed 12/5/2025.
7. Wojciech Jamroga, “Pretty Good Strategies for Benaloh Challenge”, E-Vote-ID 2023, LNCS 14230, pp.106–122, 2023; https://doi.org/10.1007/978-3-031-43756-4_7.
8. Josh Benaloh, “Verifiable Elections Tech: How Voters Can Independently Verify their Votes are Accurately Counted”, UCYB, Nov 2023, <https://youtu.be/XI9lxMETy2M>
9. <https://chain.link/education-hub/verifiable-random-function-vrf>; last accessed 9/4/2024.
10. C. Nissen et al., “Design and Evaluation of Verifiable Voting Systems Based on Tracking Code Verification”, E-Vote-ID 2024, Lecture Notes in Informatics (LNI), pp. 85-100; <https://dl.gi.de/server/api/core/bitstreams/da4452f2-240f-454c-b073-7f51ab54f3fd/content>
11. P. Ronne et al., “Expanding the Toolbox: Coercion and Vote Selling at Vote Casting Revisited”, E-Vote-ID 2024, Springer Verlag, DOI:10.1007/978-3-031-72244-8_9
12. S. Bell et al., “STAR-vote: a secure, transparent, auditable, and reliable voting system”. In: EVT/WOTE (Aug 2013)
13. Dan Wallach, lecture on STAR vote, 2017, Rice University, https://youtu.be/Cl8q_356zZs
14. Ben Adida et al., “Merge: Matching Electronic Results with Genuine Evidence for Verifiable Voting In-Person at Remote Locations”, 2024, <https://arxiv.org/abs/2410.06705>
15. Lucca Hirschi, Lara Schmid, and David Basin, “Fixing the Achilles Heel of E-Voting: The Bulletin Board”, IEEE CSF, 2021; <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9505201>; <https://github.com/LCBH/BulletinBoard-CSF21>
16. Olivier Pereira and Dan Wallach, “Clash Attacks and the STAR-Vote system”, E-Vote-ID 2017, LNCS10615, pp.228–247; https://doi.org/10.1007/978-3-319-68687-5_14.

Appendix A: Protocol Analysis

A.1 Cost of Devices & System Assumptions

The first device is assumed to function properly, [5], with a high-resolution camera, scanner or both; the second device needs a scanner to read the HBID. Both devices must be completely isolated from any network and contain a removable memory card. *We assume that the printed receipts can never be forged*, through a special paper for example or maybe some kind of automatically printed seals and/or watermarks; also, we assume *the voting machine can never print 2 sealing receipts for the same HBID*.

The Election Authority trust is controlled by the published summary reports; subtotals must be consistent with totals, and fault intervals can be identified. Finally, *the properties of HBID* are a primary assumption.

A.2 Protocol Robustness Against Known Attacks & Malicious Devices

-Malicious Device2:

Benaloh challenge and the public bulletin board are first line defenses to detect malicious voting devices, and the proposed system applies both; many other attack scenarios, like the two machines colluding together, can be detected by RLAs and consistency checks with device1 as in Fig.2.

-Bulletin board attacks

In our proposed system the BB has no need to know the voter identity before showing the results, which excludes the possibility of showing different views to different voters; i.e., the attacks in [15,16] are not applicable here, since voters cannot be deceived to check the same entry; they can't possibly print from the same device in the same booth at the same time. Finally, manipulations like adding/deleting/altering votes will contradict with subtotals in the summary reports and sealing receipts count.

-Voting Servers & Election Authority:

EA is the only one trusted to know the vote and the clear ID, that's true for the Swiss post too since it generates the codes and send them by email; in fact, if the numbers are not so large, similar codes could be used here too and printed in receipt1. Still, EA servers are not trusted alone to alter the results, since each voter has receipts and each voting location, booth, published a summary report; in addition to cryptographic functions, the summary reports contain the number of voters and the subtotals for each candidate in the clear to be readable by judges.

Malicious Entity	Defense
Device 2	Benaloh Challenge
Device 1 & Device 2	Summary reports, RLAs, consistency checks
Public Bulletin Board (PBB)	Clash attacks and alike not applicable; PBB doesn't know the checking voter (identify with time & booth)
Election Authority (EA)	Trusted in privacy, <u>voters' receipts+summary reports protect integrity</u>
Poll Officers	The linear function + auditors + RLAs

Table 1: System defense against malicious entities

-Absent Voting:

Individual voters who did not vote, or doubting parties, can start by first checking non-inclusion proofs from the summary report Verkle or Merkle value. If they found a contradiction, they can go to court; i.e., the ZKP is used only as an enough reason to start the investigation. The court then asks the authorities to bring the final receipt and the whole voter's record; there is more than one possibility:

- The authorities bring the required data accusing the voter of false claims. Then both the vote timing and the handwritten signature should be checked; for example, if the voter can prove he/she was elsewhere during this time then the vote is forged.
- The authorities claim a wrong ZKP implementation saying they agree the voter did not vote, but this doesn't reflect a problem in the election results.
 - Since the summary reports can limit the doubt only to the interval between 2 checkpoints, the court may ask for their full records to perform a limited manual check and count.
 - Also, the court may call a random sample of this interval's voters to come with their receipts for a manual consistency check, where RLAs rules may be applied.
 - Here, the worst-case court decision is to falsify the doubted intervals (not a complete election) and could vary on a case-by-case basis. The decision could depend on the number of complaining voters, the error ratio, the uncertainty of answers, ...; the court may also ask a cryptographic expert on this one (why the ZKPs gave wrong results, what are the possible reasons given the code in hand, ...) but only as an auxiliary help.

-Double Voting:

A deviation from the summary reports may happen due to detecting a repetition of the same HBID in different booths. The voting servers, by holding all the data, can locate the complete records with the time and biometric checks to trace the error and discard those votes. However, note that this situation is nearly impossible to happen; it could result only from a collision in the generated *HBID* which violates an original system assumption. If an adversary managed to get access to device2 with an old passing ticket after the voter leaves, this should be rejected by device2; it is after printing a previous sealing receipt for this HBID, and for passing the time limit between the two devices. If device2 was malicious and allowed it, this will be discovered through the

summary reports. A voter who managed to vote in two different booths, although should be rejected by device1, will get two different HBIDs.

A.3 Protocol Robustness According to Performance Metrics

-Individual verifiability is achieved through voters' receipts and the public bulletin board; every voter can search for the time recorded in his/her final receipt.

-Individual Court verifiability any voter, or a group of voters can be gathered by a losing candidate (or a winning candidate defending his/her position), to verify or falsify their true votes in front of court. The court has 3 main sources for the truth: voter's receipt, Election Authority (*EA*) receipt, and the Public Bulletin Board (*PBB*); cryptographic SNARKs from the summary reports (Merkle/Verkle of final/multiple votes) may be used only as an asserting clue.

-Soundness of court verifiability:

- A malicious voter falsely claiming he/she did not vote, will not succeed in deceiving the judges as explained in A.2 under absent voting.
- Similarly, a malicious voter cannot falsely claim a fraud vote to disrupt the election if judges can perform the simple function calculation by themselves then compare it with the *EA* receipt and *PBB* records. Even if *EA* refused to respond, judges should search the *PBB* for the (time, booth, vote count) in the voter's receipt and will not find it but will find a slightly higher time and count; the time difference would be less than the average difference between 2 voters and proportional to the difference in vote count. Hence, the judge will reject the voter's claim that this is the final receipt; a non-inclusion proof from the corresponding summary report could be used as an auxiliary assuring evidence.

-Completeness of court verifiability:

If *EA* colluded to properly alter the *BB* record of an honest voter, then judges must call *EA* to submit the corresponding sealing receipt; since we assume sealing receipts cannot be printed more than once for the same *HBID*, the truth will be revealed. If *EA* claimed losing their receipt, then it is their responsibility; they are to be assumed malicious. The court may rule at once or use a cryptographic SNARK from the corresponding summary report for assurance only.

-Universal verifiability is achieved through summary reports, biometric checks, sealing boxes count, RLAs, and zero knowledge proof queries to the server. We believe those results will be sound in court too, since the subtotals were published periodically in an immutable blockchain and handwritten signed records. Extra queries are provided by the election authority and cannot be forged for a certain booth without the collusion of both auditors and polling officers. In the worst-case scenario, poll workers receipts could be manually checked one by one against the public bulletin board data, and errors traced back with booth number and voting time.

-Universal Court verifiability if a losing candidate gathered a group of verifying voters that equals or exceeds the RLA sample size, the court can call for a manual recount if votes were proved to be wrong.

-End-to-End Verifiability:

The receipts prove votes are “casted as intended”, while the public Bulletin Board assures that votes are “recorded as casted” and consequently “counted as recorded”.

-Coercion Resistance:

The proposed system is completely robust against vote buying/coercion from external players by handing them an old receipt; even if their purpose was to *boycott* the election or choose “*reject all*”, our system provides these options as valid choices. The public Bulletin Board is not a risk here, since it does not include the ID or the HBID values. If the threat came from the poll workers, the hiding function provides a reasonable protection at election time and HBID is assumed to be irreversible (hiding); i.e., they cannot retrieve the IDs of undesired votes to trace them afterwards.

Voter's Claim	“ <i>I can' find my vote in PBB</i> ” Or “ <i>I didn't vote, but Non-inclusion Proofs says I did</i> ”
EA brings a sealing receipt and voter's whole record with signature/authentication supporting their claim	Malicious Voter
No sealing receipt supports EA claim (contradicts or does not exist)	Falsify election in this interval in this booth
EA claims lost receipt	Their responsibly, rule for voter
EA doesn't show up in court	Assumed guilty, rule for voter
EA claims “an error in non-inclusion proofs, we agree this voter did not vote”	Perform a manual recount in the specified booth & interval

Table2: Court Judgment according to different Election Authority (EA) Reactions

